Quorums

how many replicas must say yes

Phase 4 videos on YouTube

In one line: with N replicas, a write waits for W acknowledgements and a read asks R replicas; if R + W > N, every read set overlaps the latest write set, so tuning W and R trades latency and availability against consistency, and repairs bring stale replicas back up to date.

1 The stale replica problem

You store every value on three replicas. A write succeeds on two of them; the third is still missing it. Now a read arrives. If it asks only the stale replica, the user sees old data. If it waits for all three, one slow node makes every read slow. **Quorums** are the middle way.

2 N, W and R

N is the number of replicas for a key. A write is successful once **W** replicas confirm it; a read queries **R** replicas and returns the newest version among their answers (by version or timestamp). The rule to remember: if **R + W > N**, the set of replicas you read from and the set that took the latest write must share at least one replica, so the read sees that write.

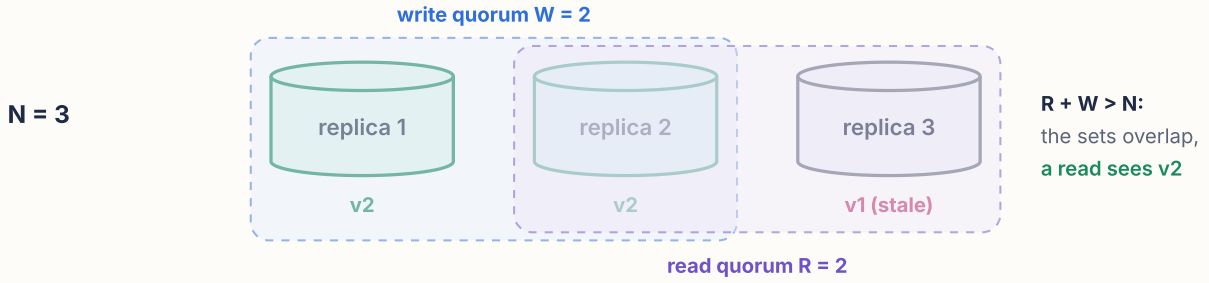


Figure 1. With N = 3, W = 2 and R = 2, any two replicas you read include at least one with the new value.

N=3, W=3, R=1	fast reads, any node down blocks writes
N=3, W=2, R=2	balanced: survives one node down for both
N=3, W=1, R=1	fastest, may read stale data (R+W ≤ N)
N=5, W=3, R=3	survives two nodes down

Figure 2. Common settings and what each one trades.

WORKED EXAMPLE: CASSANDRA QUORUM

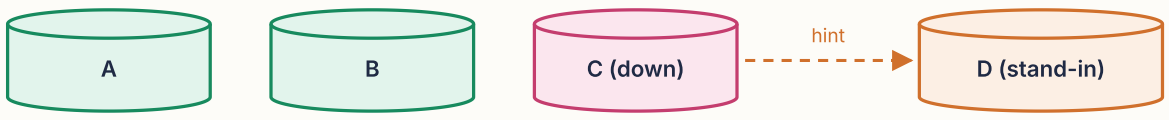
A keyspace uses replication factor 3. The app writes with consistency level `QUORUM`, so 2 of 3 replicas must acknowledge, and reads with `QUORUM` too. One replica can be down or slow and both reads and writes still succeed, and every read sees the latest acknowledged write. Writing with `ONE` and reading with `ONE` would be faster, but a read could hit a replica that hasn't received the write yet. Across data centers, `LOCAL_QUORUM` keeps the quorum inside the local data center to avoid cross-region latency.

3 Keeping replicas in sync

Quorums let some replicas fall behind, so systems repair them: **read repair** (when a read sees a stale replica, write the newest value back to it), **anti-entropy** (a background process compares replicas with Merkle trees and copies missing data), and **hinted handoff**.

4 Sloppy quorums

Dynamo-style stores can accept a write on a stand-in node when some of the key's home replicas are unreachable: a **sloppy quorum**. The stand-in keeps a **hint** and hands the data back when the home replica returns. This raises write availability, but R + W > N no longer guarantees you read the latest value until the hint is delivered.



C's write goes to D with a hint; D hands it back when C recovers

Figure 3. Hinted handoff keeps writes flowing while a replica is down.

COMMON MISTAKES

- Assuming R + W > N gives full linearizability: concurrent writes and failed writes that partially applied can still surprise you.
- Using ONE/ONE and expecting to read your own writes.
- Forgetting repairs: replicas drift and deleted data can come back (tombstones expire).
- Cross-region QUORUM on every request, paying WAN latency each time.

5 Key terms

N: number of replicas per key

Quorum: usually a majority: ⌊N/2⌋ + 1

Anti-entropy: background replica comparison (Merkle trees)

W / R: acks needed for a write / replicas asked on a read

Read repair: fix stale replicas during reads

Sloppy quorum: stand-in nodes take writes during failures

6 Interview questions

Q1. What does R + W > N guarantee? BASIC

That the replicas read and the replicas that acknowledged the latest write overlap, so a read sees the latest successfully written value.

Q2. With N = 3, how many failures can QUORUM reads and writes survive? BASIC

One. Each needs 2 of 3 replicas, so one can be down or slow.

Q3. You need very fast reads and can tolerate slower writes. Which W and R? SCENARIO

W = N and R = 1: every write reaches all replicas, so any single replica is up to date. But any one node down blocks writes.

Q4. Users in a multi-region Cassandra cluster complain about latency. What do you change? SCENARIO

Use LOCAL_QUORUM so each request waits only for replicas in the local data center, and replicate to other regions asynchronously.

Q5. Can you still read stale data with R + W > N? TRICKY

Yes: during a write in progress, after a write that failed on some replicas, with sloppy quorums, or when clocks decide the 'newest' version wrongly.

Q6. What's hinted handoff, and what's the catch? TRICKY

A stand-in node stores a write for a down replica and forwards it later. It improves availability, but until the hint is delivered a quorum read may miss that write.

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